



Interview with...

We put some questions to Dr Canan Dagdeviren, an assistant professor at the Massachusetts Institute of Technology (MIT). She works on devices that create electricity from movement.

Q: We know it is something to do with energy, but what do you actually do?

A: I create flexible devices that look like stickers and change energy from the natural movements of internal organs, such as the beating of a heart or the expanding and deflating of a lung, into electricity. Enough electricity can be made to power a machine such as a pacemaker, which is put inside the body to regulate irregular heartbeats.

Q: What made you decide to become a scientist?

A: When I was a child in the early 1990s, I learned that my grandfather had died of heart failure at just 28 years old. I made a promise to myself to do something to tackle the condition that struck my grandfather.

Q: What are these devices called?

A: They are called piezoelectric devices, and they create a small electric charge as they bend. Stuck to the lungs, for example, the device changes shape as a person breathes in and out, delivering a steady current.

Q: What are the uses of these devices?

A: They could be used anywhere we need electrical energy. Inside the body, they can power medical machines, so you don't have to change the machine's batteries.

Q: What sort of equipment do you use?

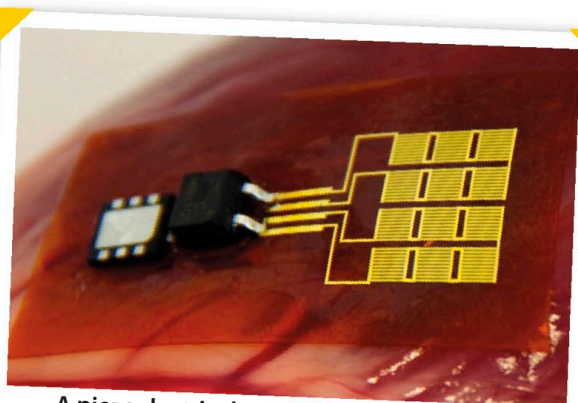
A: I use drawing programs on a computer to design the structure of my device. I also use microscopes to see the parts of the device itself, which may be very small.

Q: What are the best and worst parts of your job?

A: The best part is that I can make my dreams come true. There is no worst part. If you really love to do something, you'll experience no worst at all.

Q: What do you think the future of making energy will be?

A: Little, tiny devices will be a part of your clothing, and whenever you do your daily activities, you'll be generating electrical power. Free energy!



A piezoelectric device can make electricity using the movement of internal organs.